Session-IV: Groundwater recharge: September 1st week

Program Schedule

S.No	Time	Content	Method	Materials	Facilitator
1	10 min.	Recap.	Large group discussion	Photos, charts, markers, charts of third session.	Participants recollect the contents discussed in third session.
2	40 min.	Factors influencing recharge.	Small group discussion	Charts and markers.	Participants identify the influencing factors of groundwater recharge.
3	15 min.	Group dynamics: Cooperation among GMC members.	Game	Thread roll, soft drink bottle, pens.	Participants understand the importance of cooperation between members to accomplish tasks.
4	20 min.	HESA – Slope.	Small group discussion	Charts, Markers.	Participants discuss about slope and suggest methods to practice to if slope is more.
5	30 min.	Short Study- (A) Groundwater recharge process; and (B) factors influencing recharge process.	Demonstration	Sponge, water, glass Plastic trays, red soil, sand.	• Participants observe and discuss rate of groundwater recharge. • Participants discuss the various factors that influence groundwater recharge.
6	10 min.	Homework: Collection of rainfall data (June-October), rock types and its extent.	Small group discussion	Charts and markers.	Discuss the ways of collecting rainfall data and rock types.
7	10 min.	Evaluation.	Large group discussion	Large group discussion.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session, participants will be able to:

- Recap the previous session,
- Discuss the natural and artificial recharge methods,
- Discuss the factors influencing recharge (HESA),
- Discuss the recharge rates (GEC norms) in different rock types,
- Discuss the importance of cooperation among GMC members, and
- Discuss collection of rainfall data (June to October).

Process

1. Recap Time: 10 min.

Objective : Participants recap the topics discussed in the previous session.

Content : Recap session three contents:

- Hydrological data and its relevance,
- Measuring devices,
- Procedures of data collection, recording in HMR books and dissemination,
- Home work: Observation of water levels,
- Water levels (HESA), and
- Importance of Book- keeping.

Method : Using farmer participant as a resource person

Materials : Charts and pictures used during the previous sessions.

Process :

- Welcome the participants to the fourth session.
- Request one farmer to share the contents discussed in the third session.
- Ask the participants in large group to add any missing points in the recap.
- Briefly discuss the contents and the learning from the previous session.

2. Factors influencing groundwater recharge Time: 40 min.

Objective : Participants discuss the ways of recharge of water into soil and influencing factors of groundwater recharge.

Content : Natural and Artificial Recharge

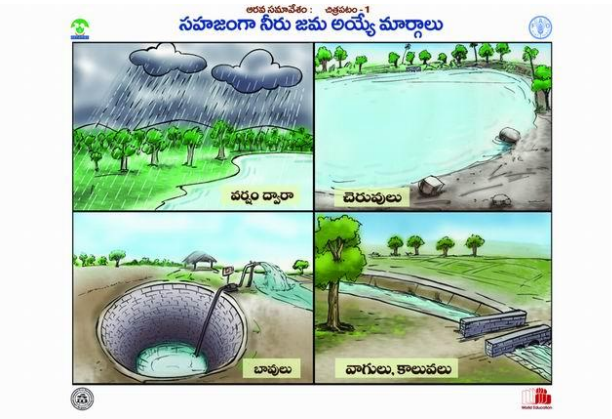
- Source of natural recharge and artificial recharge.
- Factors influencing groundwater recharge.

Method : Large group discussion

Materials : Pictures, charts and markers.

Process

- Begin a discussion with the following words. We have been discussing about groundwater since three sessions. Can you share your thoughts on how water seeps into the ground and what is the primary source?
- Write participants responses on a chart paper and identify the natural and artificial ways in which water percolates the ground.
- Use the following visuals to encourage the discussion.



Natural ways of recharge.



Artificial ways of recharge

- Summarize the discussion by stating that main source for recharge is rain, oceans and snow. Recharge can be done by building some artificial recharge structures like check dams, injection wells, contour bunds etc, but process is expensive and cannot be done on a large scale.

Factors influencing recharge

- Next, initiate the discussion by stating that till now we discussed the ways of recharge. Let us now discuss the influencing factors of recharge.
- Request participants to work in small groups to discuss the factors influencing groundwater recharge.
- Encourage participants to use the HESA frame work and present their group work.
- After presentation of all groups, write the list of factors influencing the groundwater recharge on a chart and summarize the discussion.

Tips to the facilitator

Factors influencing recharge are:

- Rainfall,
- Area, soil types,
- Rock formation, slope and drainage.

3. Group dynamics- Cooperation among GMC members Time: 15 min.

Objective : Participants participate in a game to understand the importance of cooperation between members to accomplish tasks.

Materials : Thread roll, soft drink bottle, pens.

Process

- Cut the thread into strings of three meters long.
- The number of strings should be equal to the number of participants.
- Divide the participants into groups.
- Tie five strings of equal length.

- Ask each group to have its members stand in a circle facing inwards.
- Give one pen with strings tied to it to each group.
- Ask each group member to hold the loose end of one string.
- Place the bottle of soft drink on the ground a little away from the centre of the group.
- Ask the group members to coordinate their actions to gently drop the ball pen into the soft drink bottle.
- The first group to complete the activity will be declared as the winner.
- Ask the following questions after the game:
    - What are the reasons for winning?
    - What are the reasons for not winning?
- Encourage the farmer participants to discuss the need for same kind of cooperation among GMC members.

4. HESA – Slope Time: 20 min.

- Encourage participants to discuss in small groups about the slope in the Hydrological Unit.
- Request each group to present their group work to large group.
- After completion of all group presentations summarize the discussion by sharing that the decisions should be implemented and monitored by HUN and clear action plan need to be discussed in the group.

5. Short Study **Time: 30 min.**

(A) *Groundwater recharge process* Time: 15 min.

Objective : Participants will observe and discuss rate of groundwater recharge.

Rationale : Usually farmers assume that most of the rain water infiltrates into soil. But in reality most of the rain water flows into rivers through streams and rivulets. It will be helpful for the farmers to know about the phenomenon that only a small fraction of rain water (1-12%) infiltrates into the soil. This can be explained by conducting an experiment.

Materials : Sponge, water, and glass.

Usage

- Take a dry sponge.
- Pour 100 ml. of water on sponge like rainfall.
- Collect water into a glass by squeezing the sponge. Ask the farmers whether the amount of water collected in the glass is 100 ml.

- Discuss the relative absorbing capacity of sponge with that of the soil. Explain that all the rainfall received does not infiltrate into the soil and it depends upon the rock formation and its extent.

(B) *Factors influencing groundwater recharge* Time: 15 min.

Objective : Participants will discuss the various factors that influence groundwater recharge.

Rationale : Various factors influence rate of groundwater recharge /percolation. The percolation rate depends on slope, rock formation and soil types. The following experiment will help demonstrate this.

Materials : Three plastic trays (30 x 20cm), three big plastic trays (40 x 30 cm), two types of soil (red soil, sand), bucket full of water, and mug.

Usage

- Take three plastic trays (30 x 20 cm).
- Fill two trays up to the brim with red soil.
- Fill the third tray with sand and red soil mixed in 1:1 ratio.
- Place each of these trays in a larger tray to collect water separately.
- Keep one of the two trays filled with red soil in a slant position. Keep the other two trays one filled with red soil and the other with mixed soil (red soil + sand) in a horizontal position.
- Simultaneously pour a mug of water into each tray with uniform speed.
- Measure the water collected in the three larger trays after two minutes.
- Ask the participants to observe the quantity of water collected in each of the big trays. Usually, more water is collected from the tray with slope, less water from the tray with red soil placed horizontally and even lesser water from the tray with mixed soil.

6. Homework- collection of rainfall data and rock types Time: 10 min.

Objective : Participants discuss the ways of collecting rainfall data, rock types and its extent.

Content :

- Rainfall data (June to October).
- Rock types and its extent.

Method : Large group discussion

Materials : Charts and markers.

Process

- Encourage participants to discuss the ways to collect rainfall data received during June to October and rock types in the hydrological unit area.
- Write participant responses on a chart and request participants to get the accurate data.

7. Evaluation Time: 10 min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- What are the different sources of natural recharge?
- Assume that there is equal rainfall in two different places; does the same quantity of water percolate the ground in both the places?
- Can you list the data required to estimate the groundwater recharge?
- Encourage participants to share their day's learning and thank for their participation.